

An Empirical Analysis of Hallucination Behavior in LLMs in the Medical Domain

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Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in generating fluent, contextually relevant text. However, they frequently produce hallucinations — outputs that appear plausible but are factually incorrect, unsupported, or fabricated. This problem is especially critical in high-stakes domains such as medicine, where incorrect information can have serious real-world consequences. We present a controlled empirical study of hallucination behavior across three model families (Llama-3-8B, BioMistral-7B, and BioGPT) on two medical QA benchmarks (MedQA and PubMedQA), evaluating four intervention conditions: zero-shot prompting, chain-of-thought (CoT) prompting, retrieval-augmented generation (RAG), and supervised fine-tuning (SFT). Our results show that larger instruction-tuned models benefit most from RAG, CoT prompting consistently degrades performance relative to zero-shot due to output noise and parse failures, and fine-tuning improves BioGPT accuracy by 6.66 percentage points while eliminating parse failures entirely. Combining fine-tuning with RAG, however, leads to catastrophic output degradation, suggesting that task-specific fine-tuning and retrieval augmentation are incompatible in their current forms for this model.

1 Introduction

Large Language Models (LLMs) have revolutionized natural language processing by demonstrating unprecedented capabilities in generating coherent, contextually relevant text. However, a significant challenge is their propensity to produce hallucinations — outputs that may be fluent and plausible but are factually incorrect, unsupported by evidence, or entirely fabricated. This issue is particularly concerning in medicine, where incorrect information can lead to serious consequences for patient care and safety.

Despite the growing body of research on hallucination in general NLP contexts, relatively little empirical work systematically characterizes hallucination behavior in medical LLMs, or directly compares the effectiveness of inference-time versus training-time mitigation strategies. In this paper, we address this gap by evaluating multiple LLMs on standardized medical QA benchmarks under four experimental conditions: zero-shot prompting, chain-of-thought prompting, retrieval-augmented generation, and supervised fine-tuning. By systematically comparing these interventions, we aim to answer not only how often LLMs hallucinate on medical questions, but which type of intervention works best — and whether combining them yields further gains.

2 Related Work

2.1 Hallucination in Large Language Models

Hallucination is a widely recognized problem in LLMs, where a model generates fluent and plausible text that is factually incorrect, unsupported by the input, or inconsistent with external knowledge. Prior work has categorized hallucinations into intrinsic hallucinations, which contradict the source context, and extrinsic hallucinations, which introduce information not supported by the provided evidence (Ji et al., 2023). Hallucination has also been studied in abstractive summarization, where models often generate coherent but unfaithful content (Maynez et al., 2020). These findings are relevant to medical LLMs because medical responses require both factual accuracy and faithfulness to clinical context.

2.2 Medical Question Answering and Clinical Reasoning

Recent work has shown that LLMs can achieve strong performance on medical QA benchmarks such as MedQA, MedMCQA, and PubMedQA,

but remain vulnerable to confident errors on complex clinical reasoning tasks (Liévin et al., 2024). Medical benchmarks provide a useful setting for studying hallucination because they offer verifiable ground-truth answers. Prior studies have also developed medical hallucination benchmarks, such as Med-HALT, which evaluates reasoning-based and memory-based hallucination in medical LLMs (Pal et al., 2023).

2.3 Chain-of-Thought Prompting

Chain-of-thought (CoT) prompting encourages a model to generate intermediate reasoning steps before producing a final answer (Wei et al., 2022). While CoT has been shown to improve reasoning on complex tasks, it can also produce longer generations that introduce additional opportunities for error. In medical QA, a model may generate a fluent but incorrect reasoning chain before giving its final answer, and CoT outputs may be harder to parse reliably (Wang et al., 2023).

2.4 Retrieval-Augmented Generation

Retrieval-augmented generation (RAG) combines a generative model with an external retrieval component, allowing the model to condition its output on retrieved documents rather than relying only on parametric knowledge (Lewis et al., 2020). In the medical domain, retrieval can help ground model outputs in biomedical abstracts or clinical documents, potentially reducing extrinsic hallucination. However, RAG does not fully eliminate hallucination, as a model may still misinterpret retrieved evidence or fail to align its answer with the retrieved context.

2.5 Fine-Tuning for Medical NLP

Supervised fine-tuning (SFT) on domain-specific data has been shown to improve model performance on medical tasks (Singhal et al., 2023). Domain-specific pretraining, as in BioGPT (Luo et al., 2022) and BioMistral (Labrak et al., 2024), has also been proposed as a way to reduce hallucination by internalizing medical knowledge during training. However, the relative effectiveness of fine-tuning versus inference-time interventions such as RAG remains underexplored, motivating our direct comparison.

3 Methods

3.1 Datasets

We evaluate on two medical QA benchmarks that represent different task types and hallucination detection modalities.

MedQA (USMLE) (Jin et al., 2021) consists of approximately 12,000 multiple-choice questions derived from United States Medical Licensing Examinations, covering a broad range of clinical topics. Gold-standard answer letters allow direct accuracy measurement. We evaluate on the held-out test split, sampling 450–500 items per experimental condition.

PubMedQA (Jin et al., 2019) consists of 450 expert-annotated yes/no/maybe questions derived from PubMed research abstracts. Each question is paired with the source abstract, making it ideal for evaluating context-grounded generation and retrieval augmentation. We use the full labeled training split (450 items) for all PubMedQA conditions.

3.2 Models

We compare three models spanning capability tiers:

Meta-Llama-3-8B-Instruct is a large general-purpose instruction-tuned model serving as our strong baseline. It was not fine-tuned on medical data.

BioMistral-7B (Labrak et al., 2024) is a medically-adapted model built on Mistral-7B and further pretrained on biomedical corpora. It allows us to test whether domain-specific pretraining reduces hallucination relative to a general model.

BioGPT (Luo et al., 2022) is a smaller (1.5B parameter) generative model pretrained on PubMed abstracts. It serves as both a small-model baseline and the target for our supervised fine-tuning experiments.

3.3 Experimental Conditions

We evaluate each model under the following conditions:

Zero-shot prompting provides the model with the question and answer choices (MedQA) or question alone (PubMedQA) with no additional context or reasoning instruction.

Chain-of-thought (CoT) prompting asks the model to reason step-by-step before providing a final answer, ending with `Answer: <letter>`.

Retrieval-augmented generation (RAG) uses BM25 retrieval over the PubMedQA abstract cor-

pus with top- $k=2$ abstracts prepended as context, using a fixed prompt template across all models.

Supervised fine-tuning (SFT) fine-tunes BioGPT on the full MedQA training split (10,178 samples) for 3 epochs using standard next-token prediction loss, learning rate 2×10^{-5} , effective batch size 16, and fp16 mixed precision on a single A100 GPU.

SFT + RAG applies RAG at inference time to the fine-tuned BioGPT model, testing whether the two interventions are complementary or redundant.

3.4 Evaluation Metrics

Accuracy is the proportion of outputs that match the gold-standard answer.

Error rate is the proportion of outputs that are either incorrect or unparseable. For MedQA conditions (zero-shot, CoT, SFT), we use error rate rather than “hallucination rate” since there is no retrieved context to assess faithfulness against — incorrect answers reflect factual errors rather than hallucination in the retrieval faithfulness sense.

Hallucination rate is reserved for RAG conditions on PubMedQA, where it captures outputs that are either incorrect or unfaithful to the retrieved context.

Parse failure rate is reported separately as the proportion of outputs from which no structured answer (letter or yes/no/maybe) could be extracted. Parse failures are counted as errors in all conditions since an unparseable output is clinically unusable.

BERTScore F1 measures semantic similarity between generated outputs and gold answers, serving as a secondary signal.

4 Results

4.1 RQ1: Zero-Shot and Chain-of-Thought on MedQA

Table 1 summarises the performance of Meta-Llama-3-8B-Instruct on MedQA under zero-shot and CoT prompting.

Zero-shot prompting outperforms CoT on every metric. CoT produces longer reasoning traces that introduce more formatting artifacts, repetition, and parse failures, leading to a higher measured error rate. These results suggest that explicit reasoning prompts do not automatically improve performance on medical multiple-choice QA.

We observe a consistent pattern with Qwen2.5-3B-Instruct on MedQA (200 samples), as shown in Table 2.

Metric	Zero-Shot	CoT
Correct	302	282
Accuracy	60.4%	56.4%
Error Rate	39.6%	43.6%
Parse Failures	4	18
BERTScore F1	0.819	0.814

Table 1: MedQA results for Llama-3-8B-Instruct (500 samples).

Metric	Zero-Shot	CoT
Correct	86	47
Accuracy	43.0%	23.5%
Error Rate	57.0%	76.5%
Parse Failures	1	12

Table 2: MedQA results for Qwen2.5-3B-Instruct (200 samples).

Qwen2.5-3B-Instruct performs substantially worse than Llama-3-8B-Instruct under both conditions. The gap between zero-shot and CoT is even larger for Qwen2.5-3B, suggesting that smaller models are more susceptible to the noise introduced by chain-of-thought reasoning.

4.2 RQ2: Retrieval-Augmented Generation on PubMedQA

We evaluated four models under RAG conditions on PubMedQA using BM25 retrieval with top-2 abstracts. Table 3 reports the results.

Model	Acc.	Halluc.	BERT F1
Llama-3-8B	73.4%	28.9%	0.850
BioMistral-7B	52.9%	54.9%	0.816
BioGPT	54.4%	89.1%	—
Qwen2.5-3B	19.0%	81.0%	—

Table 3: PubMedQA RAG results (450 samples for Llama-3 and BioMistral; 200 samples for Qwen2.5-3B).

Llama-3-8B-Instruct achieves the highest accuracy and lowest hallucination rate, suggesting that larger, general-purpose instruction-tuned models benefit most from retrieval augmentation. BioGPT and Qwen2.5-3B exhibit very high hallucination rates; inspection reveals that both models overwhelmingly predict “maybe,” failing to commit to definitive yes/no answers even when the retrieved evidence supports one.

4.3 RQ3: Effect of Fine-Tuning on Hallucination

We fine-tuned BioGPT on the full MedQA training split (10,178 samples) for 3 epochs. Training loss decreased steadily across epochs ($2.05 \rightarrow 1.73 \rightarrow 1.63$) with validation loss following the same trend ($1.78 \rightarrow 1.72 \rightarrow 1.71$), confirming convergence without overfitting. The fine-tuned model was evaluated on a matched subsample of 450 MedQA test items (seed=42) alongside the zero-shot baseline. Table 4 reports the results.

Condition	Accuracy	Error Rate	Parse Fail
BioGPT zero-shot	17.78%	82.22%	23.11%
BioGPT fine-tuned	24.44%	75.56%	0.00%

Table 4: RQ3: BioGPT on MedQA before and after fine-tuning (450 matched samples).

Fine-tuning improved accuracy by 6.66 percentage points and completely eliminated parse failures ($23.11\% \rightarrow 0.00\%$), indicating the model learned to produce clean answer letters consistently. Despite this improvement, fine-tuned BioGPT (24.44%) remains substantially below zero-shot Llama-3-8B-Instruct (60.4%), suggesting that model scale and instruction-tuning capability are more important factors than domain-specific fine-tuning alone.

4.4 RQ4: RAG and Fine-Tuning Interaction

To test whether RAG and fine-tuning are complementary, we applied RAG at inference time to the fine-tuned BioGPT model on PubMedQA. Table 5 reports the interaction results alongside the individual conditions for comparison.

Condition	Accuracy	Halluc.	Parse Fail
BioGPT RAG only	54.4%	89.1%	80.0%
BioGPT RAG only	54.4%	89.1%	80.0%
BioGPT SFT only	24.44%	75.56%	0.0%
BioGPT SFT + RAG	3.33%	96.67%	93.33%

Table 5: RQ4: Interaction between fine-tuning and RAG for BioGPT on PubMedQA (450 samples).

Combining fine-tuning with RAG leads to catastrophic degradation — accuracy drops to 3.33% with a 93.33% parse failure rate. This suggests that fine-tuning on MedQA (multiple-choice format) caused the model to lose the output format flexibility required for PubMedQA (yes/no/maybe format). When presented with RAG context at inference time, the fine-tuned model fails to generate

structured responses, indicating that the two interventions are incompatible in their current forms for BioGPT.

4.5 Prediction Distribution Analysis

Qwen2.5-3B-Instruct shows a strong bias toward hedging. On PubMedQA, 87.5% of its predictions were “maybe,” compared to a gold-label “maybe” rate of only 11.0%. On MedQA (zero-shot), the model exhibited a preference for option D (55.0% of predictions vs. 26.0% in the gold labels), while CoT prompting shifted the distribution heavily toward option A (60.5%). This sensitivity of the answer distribution to the prompting strategy suggests that the model’s predictions are driven more by surface-level prompt cues than by genuine medical reasoning.

5 Discussion

5.1 Effect of Model Scale

Across all conditions, model scale and instruction-tuning capability emerge as the dominant factors in hallucination reduction. Llama-3-8B-Instruct consistently outperforms both BioMistral-7B and BioGPT despite having no medical domain pre-training. This is consistent with recent findings that large general models with strong instruction tuning can match or exceed smaller specialized models on medical benchmarks (Singhal et al., 2023). Our results suggest that investing in larger, better-aligned general models may be more effective than domain-specific pretraining to reduce hallucination in medical QA.

5.2 Inference-Time vs. Training-Time Interventions

Our results reveal an important asymmetry between inference-time and training-time interventions. RAG substantially reduces hallucination for Llama-3-8B (28.9% hallucination rate) but provides limited benefit for smaller models. Fine-tuning improves BioGPT accuracy by 6.66 percentage points and eliminates parse failures, but absolute accuracy remains low. Critically, combining both interventions (SFT + RAG) leads to catastrophic failure for BioGPT, suggesting that fine-tuning on a specific answer format (multiple-choice letters) degrades the model’s ability to generalize to a different format (yes/no/maybe) even when provided with retrieved context. This incompatibility highlights the importance of evaluating

combined interventions rather than assuming they are additive.

5.3 Limitations

Several limitations should be noted. First, we do not have a zero-shot baseline on PubMedQA for direct RAG vs. no-RAG comparison on the same questions, which would strengthen our RQ2 conclusions. Second, BioGPT’s poor performance may partly reflect its lack of instruction tuning rather than fundamental model limitations — instruction-tuned variants might perform significantly better. Third, our fine-tuning used only 3 epochs; longer training or alternative objectives such as RLHF or DPO may yield greater hallucination reductions. Finally, we use BM25 retrieval with top-2 abstracts, and denser retrieval methods such as MedCPT may improve RAG quality.

6 Conclusion

We presented a controlled empirical study of hallucination behavior in medical LLMs across four intervention conditions. Our key findings are: (1) larger instruction-tuned models hallucinate less under RAG, with Llama-3-8B achieving 73.4% accuracy on PubMedQA; (2) CoT prompting consistently degrades performance relative to zero-shot due to output noise and parse failures; (3) fine-tuning BioGPT on MedQA improves accuracy by 6.66 points and eliminates parse failures; and (4) combining fine-tuning with RAG leads to catastrophic output degradation for BioGPT, suggesting the two interventions are incompatible in their current forms. These findings highlight that model scale and instruction-tuning capability are more important than domain-specific adaptation for hallucination reduction, and that combined interventions must be carefully evaluated rather than assumed to be additive.

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